

APPENDIX A PROOF OF THEOREM 1

Let us focus on the update dynamics of the boundary estimates at overlapping monitoring areas. For a local node i , the communication-driven penalty in (14) is proportional to $\|\hat{\Phi}_{i,b}^{(t)} - \bar{\Phi}_b^{(t)}\|_{\bar{\Lambda}^{(t)}}^2$. Based on the definition of the consensus target in (13a), we can rewrite the gradient of this consensus term with respect to the prediction vector $\hat{\Phi}_{i,b}^{(t)}$.

Assuming the neural networks operate in the highly over-parameterized NTK regime, the parameter updates map directly to the function space updates. The continuous-time gradient flow counterpart of the update is:

$$\frac{d\hat{\Phi}_{\text{all}}(t)}{dt} = -\eta(\mathbf{L}_\Lambda \otimes \mathbf{I}_Q) \hat{\Phi}_{\text{all}}(t) + \mathbf{g}_{\text{local}}(t), \quad (1)$$

where $\hat{\Phi}_{\text{all}} = [\hat{\Phi}_{1,b}^\top, \dots, \hat{\Phi}_{K,b}^\top]^\top \in \mathbb{R}^{KQ}$. The matrix $\mathbf{L}_\Lambda$ is the generalized Laplacian matrix of the communication graph $\mathcal{G}_c$, with off-diagonal entries $L_{ij} = -\bar{\Lambda}_{ij}$ for $(i, j) \in \mathcal{E}_c$ and zero otherwise, and diagonal entries $L_{ii} = \sum_{j \in N(i)} \bar{\Lambda}_{ij}$. Here, $\bar{\Lambda}_{ij} = \text{diag}(1/\mathbf{v}_{j,b}^{(t)})$. The term $\mathbf{g}_{\text{local}}(t)$ denotes the projection of the gradients from the local supervision loss $\mathcal{L}_{\text{sup}}^{(i)}$.

Since the graph is undirected and connected, and the variances are strictly bounded below by $\sigma_{\min}^2$ in (9), $\mathbf{L}_\Lambda$ is positive semi-definite with a single zero eigenvalue corresponding to the all-ones vector $\mathbf{1}$. The algebraic connectivity $\lambda_2(\mathbf{L}_\Lambda) > 0$.

Decomposing the dynamics into the consensus subspace (span of $\mathbf{1}$) and the disagreement subspace, the disagreement norm $V(t) = \frac{1}{2} \hat{\Phi}_{\text{all}}^\top (\mathbf{L}_\Lambda \otimes \mathbf{I}_Q) \hat{\Phi}_{\text{all}}$ evolves as:

$$\frac{dV(t)}{dt} \leq -\eta \lambda_2(\mathbf{L}_\Lambda) V(t) + \|\mathbf{g}_{\text{local}}\|_\Lambda \sqrt{2V(t)}. \quad (2)$$

Assuming the gradients of the local NLL loss evaluated at boundaries are bounded by variance σ_{local}^2 , applying Grönwall's inequality yields:

$$\limsup_{t \rightarrow \infty} V(t) \leq \frac{\sigma_{\text{local}}^2}{\eta \lambda_2(\mathbf{L}_\Lambda)}. \quad (3)$$

Translating the continuous-time bound back to discrete-time gradient descent with $\eta < \frac{2}{\lambda_{\max}(\mathbf{L}_\Lambda)}$ directly implies that the sum of weighted inconsistency at overlapping areas converges exponentially to $\mathcal{O}\left(\frac{\sigma_{\text{local}}^2}{\lambda_2(\mathbf{L}_\Lambda)}\right)$. This concludes the proof.

APPENDIX B PROOF OF THEOREM 2

We evaluate the global expected error:

$$\mathcal{R}(\hat{\Phi}) = \mathbb{E}_{\mathbf{x}} \left[\left\| \Phi(\mathbf{x}) - \sum_{i=1}^K \alpha_i(\mathbf{x}) \hat{\Phi}_i(\mathbf{x}) \right\|^2 \right]. \quad (4)$$

Since $\sum_i \alpha_i(\mathbf{x}) = 1$ and $\alpha_i(\mathbf{x}) \geq 0$, by the triangle inequality:

$$\mathcal{R}(\hat{\Phi}) \leq \sum_{i=1}^K \mathbb{E}_{\mathbf{x}} \left[\alpha_i(\mathbf{x}) |\Phi(\mathbf{x}) - \hat{\Phi}_i(\mathbf{x})|^2 \right]. \quad (5)$$

Let $p_i = \int_{\mathcal{X}} \alpha_i(\mathbf{x}) p(\mathbf{x}) d\mathbf{x}$ be the spatial measure defined by the fusion weights. We can rewrite the integral as a weighted summation of localized risks:

$$\mathcal{R}(\hat{\Phi}) \leq \sum_{i=1}^K p_i \mathbb{E}_{\mathbf{x} \sim \mathcal{X}_i} \left[|\Phi(\mathbf{x}) - \hat{\Phi}_i(\mathbf{x})|^2 \right]. \quad (6)$$

For each local region i , classical statistical learning theory based on Rademacher complexity states that for any hypothesis D_{ψ_i} mapping to $[0, M_\Phi]$, with probability at least $1 - \delta/K$, the true local risk is bounded by its empirical risk over the local dataset $\mathcal{D}_i$ (size N_i):

$$\mathbb{E}_{\mathbf{x} \sim \mathcal{X}_i} |\Phi - \hat{\Phi}_i| \leq \hat{\mathcal{R}}_S(\hat{\Phi}_i) + 2\mathfrak{R}_{N_i}(D_{\psi_i}) + 3M_\Phi \sqrt{\frac{\ln(2K/\delta)}{2N_i}}. \quad (7)$$

Additionally, because each local estimator is updated through the consensus surrogate loss $\mathcal{J}_i^{(t)}$, the localized functions deviate from purely independent functions by the boundary discrepancy term $V(\infty)$ bounded in Theorem 1. Specifically, the Lipschitz continuity of the neural mapping implies that the boundary perturbations uniformly introduce an error term bounded by $\mathcal{O}(\Delta_b)$, where $\Delta_b = \frac{\sigma_{\text{local}}^2}{\lambda_2(\mathbf{L}_\Lambda)}$.

Taking the union bound over all K nodes ensures that the localized inequality holds jointly with probability at least $1 - K(\delta/K) = 1 - \delta$. Summing these localized bounds weighted by p_i straightforwardly yields the global inequality presented in Theorem 2. This completes the proof.